



UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
Universiti Sains Malaysia, Engineering Campus

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EEM 441

Control & Instrumentation Laboratory

Experiment 4

INTERFACING FPGA TO DIP SWITCH AND SEVEN SEGMENT

Date : _____

Group No. : _____

Group Members : _____ Matrix # _____

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Experiment 4

Interfacing FPGA to DIP Switch and Seven Segment

Objectives

1. To use Verilog code as a design entry method in Altera Quartus II Software.
2. To interface a DIP switch and a seven-segment display to the FPGA.

Equipment / Software Required

1. Altera Quartus II software
 - version 9/10/12 for Cyclone II
 - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Function generator

Introduction

Seven-segment displays are widely used to show numbers. The DE2 board provides eight seven-segment displays, 18 DIP switches, four push-button switches, and 16 LEDs. These components are hardwired on the board and are each assigned to specific FPGA pins — refer to the DE2 user manual for the full pin map.

Procedure

Activity 1 — 16-bit binary counter with frequency divider

Objective: design a 16-bit binary counter driven through a frequency divider.

1. Open Quartus II and write the frequency-divider module in Listing [4.1](#) (seven), which lowers the incoming clock frequency (Device family: Cyclone II (EP2C35F672C6N) or Cyclone IV (EP4CE115F29C7N)).
2. Create a reusable symbol for seven.
3. Write the 4-bit counter module in Listing [4.2](#) (led).
4. Create a reusable symbol for led.
5. Draw the schematic in Figure [4.1](#), connecting the seven block's most significant output bit q[16] to the clock input of the led block.
6. Assign pin numbers as shown in Table [4.1](#) (refer to the DE2 user manual).

7. Use a function generator to supply the clock signal, using 1 MHz as the clock input.
8. Determine the effective input frequency reaching the led module.
9. Discuss your result based on the observed LED output display.

seven.v – frequency divider

```

1  module seven(q, clk);
2      output [15:0] q;
3      input clk;
4      reg [15:0] q;
5
6      always @(posedge clk)
7          begin
8              q = (q + 1) % 65536;
9          end
10 endmodule

```

Listing 4.1: seven.v – 16-bit frequency divider.

led.v – 4-bit counter

```

1  module led(x, clk);
2      input clk;
3      output [3:0] x;
4      reg [3:0] x;
5
6      always @(posedge clk)
7          begin
8              x = (x + 1) % 16;
9          end
10 endmodule

```

Listing 4.2: led.v – 4-bit binary counter.

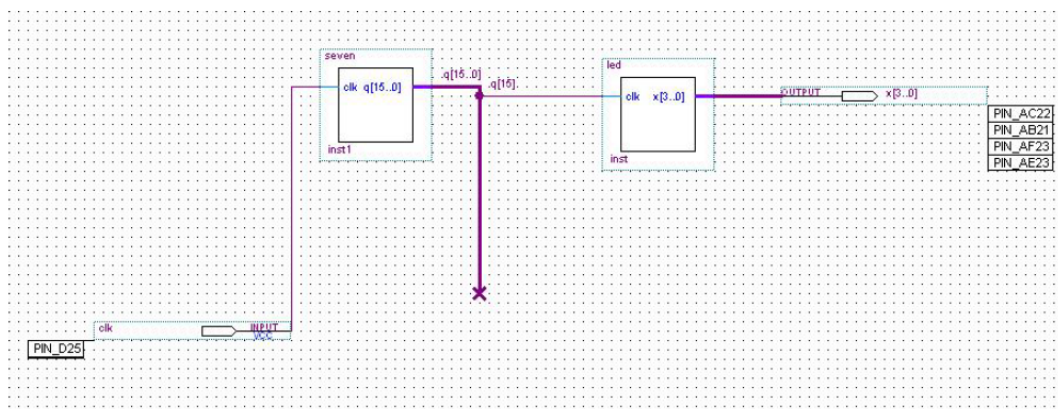


Figure 4.1: Schematic diagram for the 16-bit binary counter with frequency divider.

Table 4.1: Pin assignment for the 4-bit binary counter.

#	Node name	Direction	Pin location
1	clk	Input	
2	X[3]	Output	
3	X[2]	Output	
4	X[1]	Output	
5	X[0]	Output	

Activity 2 — 0–99 counter on seven-segment display

Objectives: (1) design a counter that counts 0 to 99; (2) display the result on the seven-segment displays.

1. Using a DIP switch as the trigger input, write Verilog code for a counter that counts from 0 to 99 once the switch is triggered.
2. Decode and display the result (tens and units digits) on two seven-segment displays.
3. Print and submit the Verilog code and simulation result.

Activity 3 — Running light

Objective: design a running-light display on the Altera DE2 board.

1. Using a DIP switch as the input, write Verilog code to implement a running-light (chasing LED) pattern.
2. Use three DIP switches to select between three different running-light patterns.
3. Print and submit the Verilog code and simulation result.